

Mathematics, 2017 EC, entrance exam, version 2

Grade 12 · Mathematics · 2017 E.C.

Time allowed: 120 minutes

Questions: 54

Total marks: 54

Name _____ Grade _____ ID No. _____ Date _____

1. If $x < 0$, then, the simplest form of $f(x) = \frac{4x+10|x|}{2x}$ is equal to: (1)
 A. 3 B. 7
 C. -3 D. -7
-
2. For the greatest integer function $f(x) = |x|$, if $|x|^2 + 5|x| + 6 = 0$, what is the value of x ? (1)
 A. $[-3,0)$ B. $[0,2)$
 C. $[-3,-1)$ D. $[-1,0)$
-
3. Which of the following intervals is the solution set of the inequality $|2 - x| < 8$? (1)
 A. $(-10,6)$ B. $(-6,10)$
 C. $(-8,8)$ D. $(-6,12)$
-
4. Which one of the following is false about the relation $R = \{(x, y) | x, y \in \mathbb{R}, y \leq -x^2 + 4 \text{ and } y \geq 2x - 4\}$? (1)
 A. Domain of R is $[-4,2]$ B. Range of R is $[-12,4]$
 C. $p(2,1)$ is a point on the graph of $R^{\{-1\}}$ D. Range of $R^{\{-1\}}$ is $[-4,4]$
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5. Which of the following is equal to $f(x) = \sqrt{(x-1)^2}$, for every $x \in \mathbb{R}$? (1)
 A. $g(x) = x + 1$ B. $g(x) = x - 1$
 C. $g(x) = |x - 1|$ D. $g(x) = |x| + 1$
-
6. The domain and range of the function $f(x) = 2x^{\frac{2}{3}}$ are respectively (1)
 A. $\mathbb{R}$ and $\mathbb{R}$ B. $[0, \infty)$ and $\mathbb{R}$
 C. $\mathbb{R}$ and $[0, \infty)$ D. $[0, \infty)$ and $[0, \infty)$
-
7. If the point (a, b) is on the graph of $y = f(x)$, which point is on the graph of $y = f^{-1}(x)$? (1)
 A. $(1/a, b)$ B. (b, a)
 C. $(-a, -b)$ D. $(a, 1/b)$
-
8. What is the equation of the tangent line to the graph of $f(x) = x^2 - x + 2$ at $(1, 2)$? (1)
 A. $x - y - 3 = 0$ B. $x - y + 3 = 0$
 C. $x - y + 1 = 0$ D. $x + y + 1 = 0$
-
9. If $f(x) = x^2 - ax + b$ and $g(x) = x^3 + c$ have the same tangent line at $p(1, 3)$, then the value of b is equal to: (1)
 A. 1 B. -2
 C. 0 D. 3

10. If $f(x) = x^2 + x + 1$, then, which one of the following is the equation of the normal line to the graph of f at $x = 1$? (1)
- A. $x + 3y + 10 = 0$ B. $3x + y + 10 = 0$
C. $x - 3y + 10 = 0$ D. $x + 3y - 10 = 0$
-
11. Let $f(x) = x^4 - 4x + 6$. For what value of x does the graph of f have a horizontal tangent line? (1)
- A. 0 B. 1
C. -1 D. 2
-
12. Let l be a line that passes through the points $P(-1, 2)$ and $Q(5, -10)$ and it is tangent to the graph of $y = f(x)$ at $x = -1$, then, $f'(-1) =$ (1)
- A. -2 B. -4
C. -5 D. 2
-
13. If $f(x) = 4x + 5$ and $(f \circ g)(x) = x^2 + 2x + 5$, then, what is the value of $g(4)$? (1)
- A. 6 B. 4
C. 29 D. 21
-
14. Given $f(x) = \sqrt{4 - x^2}$ and $g(x) = \sqrt{2x}$, which one of the following is the domain of $f \circ g$? (1)
- A. $\{x \in \mathbb{R} : -1 \leq x \leq 2\}$ B. $\{x \in \mathbb{R} : 0 \leq x \leq 2\}$
C. $\{x \in \mathbb{R} : x \leq 0\}$ D. $\{x \in \mathbb{R} : x \geq 0\}$
-
15. If $f(x) = 3 + \frac{1}{e^x}$, then, $f^{-1}(e^2 + 3) =$ (1)
- A. $1/e$ B. 2
C. -2 D. 1
-
16. Let $f(x) = \frac{1-3x}{x-2}$. Then, what is the range of $f(x)$? (1)
- A. $\mathbb{R} \setminus \{2\}$ B. $\mathbb{R}$
C. $\mathbb{R} \setminus \{-3\}$ D. $\mathbb{R} \setminus \{1/3\}$
-
17. If f is the greatest integer function and g is an absolute value function, then, what is the value of $(f \circ g)\left(\frac{7}{2}\right) + (g \circ f)\left(-\frac{10}{3}\right)$? (1)
- A. 6 B. 7
C. 0 D. 8
-
18. Let $f(x) = x^{\frac{1}{m}}$ and $g(x) = x^{\frac{1}{n}}$ be power functions where m and n are even natural numbers and $m > n$. Which one of the following is true about f and g ? (1)
- A. If $0 \leq x \leq 1$, $f(x) \leq g(x)$ B. If $x \geq 1$, $f(x) \geq g(x)$
C. If $0 \leq x \leq 1$, $f(x) \geq g(x)$ D. The domain of f and g is $\mathbb{R}$
-
19. Let $f(x) = \frac{x-1}{x+1}$ and $g(x) = \sqrt{3-2x}$ be two functions. Which one of the following is false about the combination values of f and g at $x = -3$? (1)
- A. $(f \circ g)(-3) = 6$ B. $(f/g)(-3) = 2/3$
C. $(f + g)(-3) = 5$ D. $(f - g)(-3) = -5$
-
20. If $x < 0$ and $f(x) = |x|$, then, $x - f(f(x))$ is equal to (1)
- A. 0 B. $2x$
C. $-2x$ D. $x + x^2$

21. Which one of the following is the sum of the series $\sum_{n=1}^{\infty} \left(\frac{3^n + 4^n}{6^n} \right)$? (1)
- A. 0
B. 5/4
C. 3
D. ∞
-
22. If $2^k \times 2^{k^2} \times 2^{k^3} \times \dots \times 2^{k^n} \times \dots = \frac{1}{8}$, then, $k =$ (1)
- A. 2/3
B. 1
C. -3/4
D. 3/2
-
23. If $\sum_{n=1}^k \left(\frac{n-13}{2} \right) = -25$, then, the value of $k =$ (1)
- A. 3
B. 4
C. 7
D. 20
-
24. Let $G_n = r^n$ be a geometric progression with a common ratio r . which one of the following is false about the sequence G_n ? (1)
- A. If $|r| < 1$, as $n \rightarrow \infty$ G_n converges to 1
B. If $r \leq -1$ as $n \rightarrow \infty$ G_n is vibrant
C. If $r = 1$ as $n \rightarrow \infty$ $G_n \rightarrow 1$
D. If $r > 1$ as $n \rightarrow \infty$ $G_n \rightarrow \infty$
-
25. What is the simplified form of $\frac{a^{-4} - b^{-4}}{a^{-2} + b^{-2}}$? (1)
- A. $(b^2 - a^2)/(ab)^2$
B. $(ab/(b-a))^2$
C. $((b-a)/(ab))^2$
D. $(ab)^2/(b^2 - a^2)$
-
26. Which one of the following is true about the function $f(x) = \frac{x^3 - x}{x^2 - x}$? (1)
- A. f has a vertical asymptote
B. $x = -1$ is zero of f
C. f has a horizontal asymptote
D. $x = 1$ is zero of f
-
27. If $p(x) = 3x^2$ and $q(x) = x^2 + x$, then, what is the solution set of $\frac{p(x)}{3q(x)} = \frac{1}{x} - \frac{1}{q(x)}$ (1)
- A. $\{-1, 0\}$
B. $\{1\}$
C. $\{0, 1\}$
D. $\{\}$
-
28. The value(s) of x where the graph of the function $y = \frac{x^3 - 1}{x^4 + 1}$ crosses its horizontal asymptote is(are): (1)
- A. $x = 1$
B. $x = -1$ and $x = 1$
C. $x = 0$
D. $x = -1$
-
29. If $\frac{1-x}{x^3+x} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$ then, $A + B + C =$ (1)
- A. -1
B. 0
C. 1
D. -2
-
30. Which one of the following functions has no vertical asymptote? (1)
- A. $f(x) = (x^2 - 1)/(x^2 - x)$
B. $f(x) = x/(x^3 + x)$
C. $f(x) = 4/(3x + 1)$
D. $f(x) = x^{\{-3\}}$
-
31. Which of the following is false about the graph of $f(x) = \frac{x^2 - 5x + 6}{x^2 - 4}$? (1)
- A. The graph has a hole at $x = 2$
B. The vertical asymptotes of the graph are $x = 2$ and $x = -2$
C. The horizontal asymptote of the graph is $y = 1$
D. The graph has y -intercept at $(0, -3/2)$

32. Let A and B be 3×3 matrices such that $A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & -1 & 2 \end{bmatrix}$ and $|B| = 1/2$. Which of the following is equal to $|AB^{-1}|$? (1)
- A. 1
B. 4
C. 10
D. $5/2$
-
33. If A is 3×3 matrix with $|A| = 1/2$, then, $|2 \text{ adj } A| =$ (1)
- A. 1
B. 2
C. 4
D. 8
-
34. Let A and B be 3×3 matrices with $|A| = m$, and $|B| = 1/n$, $m \neq 0$, $n \neq 0$. What is the value of $\det(A^T B^{-1})$? (1)
- A. m/n
B. mn
C. n/m
D. $2mn$
-
35. If $A = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$ then, A^{-1} is equal to (1)
- A. $\begin{bmatrix} 2 & -1 \\ -5 & 3 \end{bmatrix}$
B. $\begin{bmatrix} -2 & 1 \\ 5 & -3 \end{bmatrix}$
C. $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$
D. $\begin{bmatrix} -2 & -1 \\ -5 & -3 \end{bmatrix}$
-
36. Let $A = \begin{bmatrix} 2 & x+y \\ 6 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 8 \\ 6 & x-y \end{bmatrix}$. If $A = B$, then $x^2 - y^2 =$ (1)
- A. 32
B. 36
C. 4
D. 48
-
37. What is the sum of the series $\sum_{n=3}^{101} (-1)^n \left(\frac{1}{n} + \frac{1}{n+1} \right)$? (1)
- A. $35/101$
B. $35/101$
C. $35/101$
D. $35/102$
-
38. If the fourth term of an arithmetic sequence is -1 and the tenth term is 35, what is the sum of the first 20 terms? (1)
- A. 1520
B. 1420
C. 860
D. 760
-
39. If the sum of the first n-terms of an arithmetic progression is $S_n = n + 2n^2$, which of the following is the 21th term (a_{21}) of the sequence? (1)
- A. 903
B. 820
C. 83
D. 63
-
40. If x, x+2, and x+6 are consecutive terms of a geometric progression, then the value of x is equal to (1)
- A. 1
B. 2
C. 3
D. 4
-
41. Which one of the following is a false statement? (1)
- A. $\sum (a_i - b_i) = \sum a_i - \sum b_i$
B. $\sum (a_i \times b_i) = \sum a_i \times \sum b_i$
C. $\sum (c a_i) = c \sum a_i$
D. $\sum_{i=1}^n c = n c$
-
42. If a, b, c, d are the four geometric means between $1/3$ and 81, what is the common ratio of the geometric progression $1/3, a, b, c, d, 81$? (1)
- A. $\sqrt{3}$
B. $1/3$
C. 3
D. 9

43. If $\{a_n\}$ is an arithmetic sequence whose third term and the tenth term are 3 and 24 respectively, then, the 50th term of the sequence, a_{50} , is equal to (1)
- A. 156
B. 150
C. 144
D. 141
-
44. A single square is made from 4 matchsticks. To make two squares in a row takes 7 Matchsticks, while three squares in a row takes 10 matchsticks. How many matchsticks are in a row of 90 squares? (1)
- A. 211
B. 271
C. 291
D. 298
-
45. $\sum_{i=1}^{80} (\log_3^{i+1}) =$ (1)
- A. $1/81$
B. 4
C. -4
D. 3
-
46. The fifth and tenth terms of a geometric sequence are 640 and 20 respectively, then, the common ratio of the sequence is equal to: (1)
- A. 2
B. $1/2$
C. $-1/2$
D. -2
-
47. Which one of the following is the fractional form of the recurring decimal 0.23? (1)
- A. $23/100$
B. $23/90$
C. $23/99$
D. $23/900$
-
48. If $\sum_{n=1}^{\infty} a_n$ is a convergent series such that $\sum_{n=1}^{\infty} (a_n - 3/4^n) = 12$, then $\sum_{n=1}^{\infty} a_n$ is equal to: (1)
- A. 11
B. 112
C. 13
D. 14
-
49. A certain meeting hall has n rows of seats and 120 total numbers of seats. There are 3 seats in the first row, 5 seats in the second row, 7 seats in the third row, and so on. How many seats are there in the last row (n th row) of the hall? (1)
- A. 21
B. 23
C. 32
D. 41
-
50. If $f(x) = \sqrt{x}$, then $f'(4)$ is equal to: (1)
- A. $1/4$
B. $1/2$
C. 0
D. $1/(2\sqrt{x})$
-
51. Let $f(x) = 6x/(x+a)$. For what value of a is $f'(2a)=2$? (1)
- A. $1/3$
B. $2/3$
C. $3/2$
D. 3
-
52. If $h(x) = f(x)/g(x) + f(x)g(x)$, $f(1)=3$, $f'(1)=-2$, $g'(1)=4$, $g(1)=2$, then $h'(1)$ is equal to: (1)
- A. 4
B. -10.4
C. 5
D. 16
-
53. If $g(x) = f(x^2+4x)$, $f(-3)=6$, $f'(-3)=-5$, then $g'(-1) =$ (1)
- A. -2
B. -10
C. -30
D. -5

54. If $f(x) = 4 + |x+6|$ for all x , then what is the value of $f'(x)$ at $x = -6$?

(1)

A. -1

B. 4

C. 1

D. does not exist

— END OF EXAMINATION —

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